

Phase 2:Requirement Analysis

SmartLenderA:I-PoweredCreditCardApprovalSystem

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TeamID : SWTID-2026-4551

Role	Team Name
Lead	KarriRohith Kamesh
Member	1 SatyaSai Prasad Vithanala
Member	2 Sagiraju Ravi Eswar
Member	3 Velduti Sai Kishore
Member 4	Pampana Venkata Pradeepi

Functional Requirements

ID	Requirement
FR-1	Accept applicant details through a web form (income, employment, credit history, etc.)
FR-2	Preprocess and encode inputs the same way as the training pipeline
FR-3	Return an Approve / Reject prediction with a confidence score
FR-4	Allow retraining the model on updated historical data
FR-5	Support cloud deployment through IBM Watson Machine Learning

Non-Functional Requirements

ID	Requirement
NFR-1	Prediction response time under 2 seconds
NFR-2	Web form usable without technical training
NFR-3	Model retraining reproducible from raw data to deployment
NFR-4	System available for real-time, concurrent predictions

Data Requirements

Historical applicant records with demographic and financial details (income, employment length, existing credit, education, family status) are used together with a payment history log that is used to derive the approval label.

Hardware & Software Requirements

Python 3.10+, pip

8 GB RAM minimum for local training

Web browser for the Flask front end

IBM Cloud account (only required for Watson ML deployment)